MILITARY SPECIFICATION

TEST SET, SUBASSEMBLY MX-8047/TPM-22A

(HIGH VOLTAGE TESTER)

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Test Set Subassembly MX-8047 TPM-22A (microwave chassis and klystron power supply test set) hereinafter referred to as high voltage tester, a part of Radar Test Set AN/TPM-22A. The microwave chassis and klystron power supply test set tests microwave chassis assembly 2A1 and klystron power supply 2A2 of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Armament Research and Development Center Attn DRSMC-QA Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MIL-P-14232	- Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-C-46357	- Computer, Gun Direction; M18 Manufacturing Process and Inspection of
MIL-P-50321	- Klystron Power Supply: 11834449
MIL-P-50323	- Klystron Beam Power supply : 11834450

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-454	General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

2.1.2 Other government documents, drawings, and publications . The following other government documents, drawings, and publications form-a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9333062	- Test Set, Subassembly MX-8047/TPM-22A
9333080	- Test Set Subassembly MX-8047/TPM-22A Schematic

PACKAGING DATA SHEET

9333062	- Packaging of Test Set, Subassembly MX-8047/TMP-22A

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(Copies of specifications, standards, handbooks, drawings and publications required by the manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Fabrication. The high voltage tester shall be manufactured in accordance with Drawing 9333062 and drawings pertaining thereto. Subassemblies: shall have met the requirements of the following specifications:

Subassembly	Specification
Klystron Power Supply: 11834449	MIL-P-50321
Klystron Beam Power Supply: 11834450	MIL-P-50323

3.2 General specification. Where applicable, the high voltage tester shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishes

d. Part identification and marking

e. Workmanship

f. Soldering electrical connections

3.2.2 MIL-C-46357

a. Etched circuit assemblies

b. wiring

c. Threaded fasteners

d. Marking

e. Finishes

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3.3 Environmental service conditions.

3.3.1 Vibration. The high voltage tester shall be mounted on the vibration tabtable in normal service position. The high voltage tester shall meet the requirements of 3.4 and 3.5 after-being subjected to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and anplitude as follows:

Frequency Range	Double Amplitude
5 to 15 Hz	0.06 inch
15 to 25 Hz	0.04 inch
25 to 55 Hz	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second (1 cps/see) . Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. The high voltage tester shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes . The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds. The G value of each impulse shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds.

3.3.3 Altitude.

3.3.3.1 0 Operatin The high voltage tester shall meet the requirements of 3. and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg) .

3.3.3.2 Non-operating. The high voltage tester shall meet the requirements of 3.5 after exposure for not less than.10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg) .

3.3.4 Temperature .

3.3.4.1 Operating The high voltage tester shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +90°F.

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3 .3.4.2 Non-operating (storage) The high vol tage tester shall not suffer permanent. damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours, and +160°F for a period of 48 hours (after temperature stablization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The high voltage tester shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram, 9333080.

3.5 Performance

3.5.1 MICROWAVE CHASSIS TEST section 2A1.

3.5.1.1 Interlock indicator lamp (DS7) and interlock circuit. The interlock indicator lamp shall be illuminated when 24 V is applied between J8-V(+)* and J8-S(Ret) and switch S9 is energized in the manual service position. Interlock circuit shall be closed and continuity shall exist between J8-S and TB2-4 under the following conditions:

a. 24V applied as stated above

b. 2A1/2A2 switch in 2A1 position

c. J2/J5 door interlock switch S7 in normally open ** position

d. Interlock switch S9 energized in manual (service) or operate position.

e. Interlock plate 10548925 properly installed on subassembly A1.

* The information in parenthesis refers to the polarity applied to this terminal. This notation shall be used throughout this specification.

** Normally closed and normally open positions of all pushbutton indicator-type switches reflect the schematic designation of the two switch positions as marked on body of switch.

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3.5.1.2 Blower motor (Bl). Blower motor Bl shall operate under conditions stated in paragraph 3.5.1.1 and 115 Vac, 400 Hz is applied between J8-L and J8-M.

3.5.1.3 POWER switch (S4).

3.5.1.3.1 OFF legend. legend of S4 shall be illuminated when S4 is in normally closed position and conditions stated in 3.5.1.1 are met.

3.5.1.3.2 STANDBY legend. STANDBY legend of power indicator lamp S4 shall be illuminated when S4 is in normally open position, 24 Vdc is applied between J8-R(+) and J8-S (Ret) and conditions stated in 3.5.1.1 are met.

3.5.1.4 Output voltages (Jl) . Under conditions listed in a. through e. below, output voltages at J1 shall be as listed in f. through j. below:

a. Controls and conditions in 3.5.1.3.2.

b. Circuit breaker CB1 closed (White band not visible) .

c. 28 Vdc ± O.1V applied between J8-A/P(+) and J8-B/N (Ret) .

d. 33 Vdc ± O.1V applied between J8-T(-) and J8-U (Ret) .

e. 15 Vdc ± O.1V applied between J8-E(+) and J8-F (Ret) .

f. 28 ± O.1V between J1-1(+) and J1-2 (Ret) .

g. 28 ± O.1V between J1-3(+) and J1.-2 (Ret) .

h. 28 ± O.1V between J1-5(+) and J1-6 (Ret) .

i. 33 ± 0.1V between J1-7(-) and J1-8 (Ret) .

j. 0 to NMT 7.4V (variable by operation of Rl) between J1-A2 (+) and J1-4 (Ret).

3.5.1.5 RADIATE/STANDBY switch (S5)

3.5.1.5.1 STANDBY legend. STANDBY legend of S5 shall be illuminated when S5 is in the normally closed position and conditions in 3.5.1.3.2 are met.

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3.5.1.5.2 STANDBY mode. A voltage of 4.3 $^{+0}_{-0}$ kVdc shall be available between J5-C(+) and J5-E(Ret) , under the following conditions :

a. Conditions in 3.5.1 .5.1 are met.

b. P2 plugged into J5.

c. Circuit breaker CB1 closed (white band not visible) for more than 20 seconds,

3.5.1.5.3 RADIATE legend and radiate mode. Radiate legend of S5 shall be illuminated and an output pulse as described g. and h. below shall be available between J5-E(-) and chassis ground (E4) under the following conditions:

a. Controls and conditions are met in accordance with 3.5.1.4 for more than 2 minutes ± 12 seconds, (Delay period vesrifies proper operation of time delay relay K1.)

b. S5 in normally open position.

c. Circuit breaker CHl closed.

d. P2 plugged into J5.

e. Shorting wire between K5 normally open and normally closed contacts.

f. A pulse having the following characteristics applied to J9:

width	=	0.5 usec
amplitude	=	20 ± 2 volts peak
repetition rate	=	20 kHz ± 1 kHz
rise time	=	less than 0.05 nsec

g. Not greater than -14 kVdc at most negative point.

h. NLT 4.0 and NMT 5.0 kV amplitude from leading edge baseline to most positive point excluding positive overshoot.

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3.5.1.6 KLYSTRON FILAMENT CURRENT meter (M1) . Meter M1 shall indicate 1.5A ±10% for 20 ±2 seconds after start of operation as stated in 3.5.1.5.2 and NLT 2.0 and NMT 2.8A after 20 ±2 seconds of operation.

3.5.1.7 2A1 OVERLOAD indicator (S6)

3.5.1.7.1 High voltage power supsupply overload. HIGH VOLTAGE POWER SUPPLY legend of S6 shall be illuminated when 28 Vdc is present between P1-1 and chassis ground.

3.5.1.7.2 Modulator overload. MODULATOR legend of S6 shall be illuminated when 28 Vdc is present between P1-3 and chassis ground.

3.5.1.8 2A1 OVERLOAD reset switch (S6). Between Pl-2(+) and chassis ground (Ret) 28 ±0.3 Vdc shall be present when conditions are as stated in 3.5.1.3.2 and S6 is actuated (depressed)

3.5.2 KLYSTRON POWER SUPPLY TEST section 2A2

3.5.2.1 Interlock indicator lamp (DS7) and interlock circuit. The interlock indicator lamp DS7 shall be illumnated when 24V is applied between J8-V(+) and J8-S(Ret) and switch S9 is energized in the manual (service) Position. Interlock circuit shall be closed and continuity shall exist between J8-S and TB2-4 under the following conditions:

a. 24V applied as stated above

b. 2A1/2A2 switch in 2A2 position

c. J2/J5 door interlock switch S7 in normally open position.

d. interlock switch S9 energized in manual (service) or Operate position.

e. interlock plate 10548925 properly installed on subassembly A1.

f. shorting wire connected between S11 normally open and normally closed contacts.

9. shorting wire connected between J3-9 and J3-10.

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3.5.2.2 POWER switch (S1)

3.5.2.2.1 OFF legend. OFF legend of POWER switch S1 shall be illuminated when S1 is in the normally closed position and conditions are as stated in 3.5.2,1.

3.5.2.2.2 STANDBY legend. STANDBY legend of POWER switch S1 shall be illuminated and 28 ± 0.1 Vdc shall be between J3-A3 (+) and J3-A2 (Ret) when S1 is in normally open position and conditions are as stated in 3.5.2.1 and 28 Vdc is applied between J8-A/P(+) and J8-B/N(Ret) .

3.5.2.3 OPERATE/STANDBY switch (S2)

3.5.2.3.1 STANDBY legend and standby mode. KLYSTRON POWER SUPPLY TEST section of the high voltage tester is in the standby mode and STANDBY legend of S2 is illuminated when POWER switch S2 is in STANDBY condition (NC Position) as stated in 3.5.2.2.2. J3-A4 should read zero volts.

3.5.2.3.2 OPERATE legend and operate mode. KLYSTRON POWER SUPPLY TEST section of the high voltage tester is in the operate mode, OPERATE legend of S2 shall be illuminated 28 ± 0.1 Vdc shall be available between J3-A4 (+) , J3-A3 (+) and J3-A5 (Ret), 24 ± 0.1 Vdc between J3-4(+) and J3-A2, and, -9.0 $^{+0}_{-0}$ kVdc shall be available between J2 and chassis ground under the following conditions:

a. controls and conditions met in accordance with 3.5.2.2.2.

b. S2 in normally open position.

c. 185 volt peak-to-peak symetrical square wave applied between J3-13 and J3-14.

d. switch S10 in Test B position.

3.5.2.4 KLYSTRON POWER SUPPLY CURRENT meter (M2)

3.5.2.4.1 Standbv current. M2 shall indicate 2 amperes when conditions are as state in 3.5.2.2.2 and 14-ohm load is between J3-A3 and. J3-A2.

3.5.2.4.2 Operate current. M2 shall indicate 4 amperes when conditions are as stated in 3.5.2.3.2 and 14-ohm load is between:

a. J3-A3 and J3-A2

b. J3-A4 and J3-A5

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3.5.2.5 2A2 OVERLOAD indicator (S3)

3.5.2.5.1 High voltage power supply overload. When 28 Vdc is present between J3-1 and J3-16, HIGH VOLTAGE POWER SUPPLY legend of S3 shall be illuminated.

3.5.2.5.2 Modulator overload. When 28 Vdc is present between J3-3 and J3-16, MODULATOR legend of S3 shall be illuminated.

3.5.2.6 2A2 OVERLOAD reset switch (S3). Between J3-2(+) and J3-16 (Ret) 28 Vdc shall be present when S3 is actuated and conditions are set in accordance with any of the following:

a. 3.5.2.2.2

b. 3.5.2.3.1

c. 3.5.2.3.2

3.5.2.7 TEST A/B switch (S10). Continuity exists between P2-C and chassis ground when conditions are as stated in 3.5.2.3.2 and S10 is in TEST A position.

3.5.2.8 VOLTAGE divider (A3) . With conditions as specified in 3.5.1.5.3, the output at. J6 shall be as follows:

a. Minus 13.5 i 1V at the most negative point.

b. Plus 4.5 ±0.5V from leading edge baseline to most positive point, excluding overshoot.

c. Pulse shall be in accordance with Figure 2.

3.5.2.9 EXTERNAL BEAM POWER SUPPLY test setup. With conditions as specifie in 3.5.1.3.2, a shorting wire between TP6 and TP7, and a shorting wire between J12-E and J12-A the output across a 100 ohm resistor inserted between J17-C and J12-B shall be as follows:

a. S12 closed. A square wave with: Amplitude: 195 ± 10V peak to peak; Frequency: 3.5 ± 1 kHz; Ripple on peaks: less than IV peak to peak (not including high frequency spikes).

b. S12 open. Same as a. except: Amplitude: 97.5 ± 5V peak to peak.

c. Output of TP2 (white) with respect to TP4 (black) should read between NLT 9.OV to NMT 12.4 Vdc.

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d. Output of TP5 (blue) with respect to TP4 (black) should read between NLT 1.6V to NMT 2.2 Vdc.

e. Output of TP3 (green) with respect to TP4 (black) should read between NLT 34V to NMT 38 Vdc.

3.6 First article. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (See 4.3 and 6.2).

3.7 Workmanship. The requirements of MIL-STD-454 and other applicable specifications shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Unless otherwise specified in the item detail specification, or contract, the following types of inspection shall be conducted on the unit or product:

a. First article inspection (4.3)

b. Quality conformance inspection (4.4)

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting officer (see 6.2) for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 3 test sets.

4.3.2 Inspection to be performed. First article samples may be subjected by the Government to any or all of the examinations and tests specified in 4.4.2 of this specification and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.4.2 Examinations and tests. Except where otherwise specified herein, components and subassemblies shall be inspected in accordance with the “Inspection provisions for components and subassemblies” and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I and II and Methods of Inspection in 4.7. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.Workmanship shall be in accordance with MIL-STD-454 and other applicable specifications .

4.4.3 Rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittence of rejected lots shall be in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.5 Special Sampling. One high voltage tester out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table II.

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.4.3) .

4.6 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the referenced specifications and standards cited in 4.4.2.

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TABLE I. Classification of defects
(100 percent inspection)

	Characteristic	Requirement	Procedure
101.	Electrical	3.4	4.7.2
102.	Interlock indicator lamp (DS7)	3.5.1.1	4.7.3.1.1
103.	Blower Motor (B1)	3.5.1.2	4.7.3.1.2
104.	POWER switch (S4)	3.5.1.3	4.7.3.1.3
105.	Output voltage (J1)	3.5.1.4	4.7.3.1.4
106.	RADIATE/STANDBY switch (S5)	3.5.1.5	4.7.3.1.5
107.	KLYSTRON FILAMENT CURRENT meter (M1)	3.5.1.6	4.7.3.1.6
108.	2A1 OVERLOAD indicator (S6)	3.5.1.7	4.7.3.1.7
109.	2A1 OVERLOAD reset switch (S6)	3.5.1.8	4.7.3.1.8
110.	Interlock indicator lamp (DS7)	3.5.2.1	4.7.3.2.1
111.	POWER switch (S1)	3.5.2.2	4.7.3.2.2
112.	OPERATE/STANDBY switch (S2)	3.5.2.3	4.7.3.2.3
113.	KLYSTRON POWER SUPPLY SUPPLY CURRENT meter (M2)	3.5.2.4	4.7.3.2.4
114.	2A2 OVERLOAD indicator (S3)	3.5.2.5	4.7.3.2.5
115.	2A2 OVERLOAD reset switch (S3)	3.5.2.6	4.7.3.2.6
116.	TEST A/B switch (S10)	3.5.2.7	4.7.3.2.7
117.	VOLTAGE divider (A3)	3.5.2.8	4.7.3.2.8
118.	EXTERNAL BEAM POWER SUPPLY test setup	3.5.2.9	4.7.3.2.9

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TABLE II. Special sampling

Characteristic	Requirement	Procedure
301. Fabrication	3.1	Applicable dwg. visual
302. General specification	3.2	MIL-F-13926 visual
303. Vibration	3.3.1	4.7.1.1
304. Shock	3.3.2	4.7.1.2
305. Altitude - operating	3.3.3.1	4.7.1.3
306. Altitude - non-operating	3.3.3.2	4.7.1.3
307. Temperature	3.3.4	4.7.1.4
308. Humidity - non-operating	3.3.5	4.7.1.5

Inspection equipment shall be as listed in Table III. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See 6.3 of this specification.

TABLE III. Inspection equipment list.

Item	Description
1. Power supply	28 Vdc ±. 1V, 10A
2. Power supply	24 Vdc ±.1V, 5A
3. Power supply	33 Vdc ±.1V, 5A
4. Power supply	15 Vdc ±.1V, 01A
5. Power supply	115V ±10V, 400 Hz ± 5Hz, 200ma
6. Voltmeter	Range - 0 to 40 V, accuracy ±0.01%

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7.	Resistive load	12 ohms ±1%, 150 watts
8.	Resistive load (2 required)	14 ohms ±1%, 100 watts
9.	Ohmmeter	Standard measuring equipment to indicate continuity or infinity
10.	Voltmeter	H.V. dc, 4 to 15 kVdc electrostatic deflection, accuracy of 2%
11.	Pulse generator	Capable of providing a 0.5 usec, 20 kHz, 20 ±2 volt pulse peak to peak into 50 ohms
12.	Oscilloscope	input. sensitivity = 5 V per division Sweep period = 1.0 used per division Input resistance = 10 megohms Vertical passband = dc to 10 mHz Input capacitance = less than 5 pF Dual. Channel
13.	Oscilloscope probe	Voltage attenuation - 1000/1 Compatible with oscilloscope, item 12.
14.	Resistive load	100 ohms ±10%, 160 watts
15.	Shorting wire	Wire, insulated
16.	Shorting wire	Wire, insulated

4.7 Methods of inspection.

4.7.1 Environmental. Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with “Test Facilities and Apparatus” requirements of MIL-STD-810.

4.7.1.1 Vibration. The test shall. be performed as specified in Method 514, Procedure I, Category I, of MIL-STD-810 as modified by 3.3.1 herein. After vibration, the high voltage tester shall be tested as given in the sub-paragraph of 4.7.2 and 4.7.3 for compliance with 3.3.1.

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age tester shall be tested as given in the sub-paragraphs of 4.7.2 and 4.7.3 for compliance with 3.3.1

4.7.1.2 Shock. This test shall be performed as specified in ProcedureI,Method 516 of MIL-STD-810 except that s shock pulses shall be 11.0 ms and 15 g's, After the shock impulses, the high voltage tester shall be tested as specified in 4.7.2 and 4.7.3 for compliance with 3.3.2.

4.7.1.3 Altitude (non-operating and operating) . This test shall be performed as specified in Procedures II and III. Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.7.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating) . This test shall be performed as specified in Procedures I and III, Method 501, of MIL-STD-810 with the constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating) . This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating) . This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, the checkout shall be performed at the end of the tenth cycle, and at standard ambient conditions.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. All. following tests with the exception of 4.7.3.1.6,4.7.3.1.7, and 4.7.3.2.5 shall be performed with +28 Vdc, +24 Vdc, and 115 Vac applied to J8 using items 1, 2, and 5 as shown in Figure 1,

4,7.3.1 MICROWAVE CHASSIS TEST section, 2A1.

4.7.3.1.1 Interlock indicator lamp (DS7) Apply voltages to J8 as specified in 4.7.3. Under the conditiotis stated in 3.5.1.1a. through e., verify that continuity exists between J8-S and TB2-4. If any one of the conditions are not met, verify that continuity does not exist. Verify that DS7 is illuminated when S9 is pulled out in manual position and extinguished when S9 is pushed in or returned to its center (OFF) position.

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4.7.3.1.2 Blower motor (B1). Apply voltages to J8 as specified in 4.7.3. Under conditions stated in 3.5.1.2, verify that B1 operates.

4.7.3.1.3 POWER switch (S4).

4.7.3.1.3.1 OFF legend. Apply voltages to J8 as specified in 4.7.3. Under conditions as stated in 3.5.1.3.1, verify that "OFF" legend of S4 is illuminated.

4.7.3.1.3.2 STANDBY legend. Apply voltages to J8 as specified in 4.7.3. Under conditions stated in 3.5.1.3.2, operate S4 to the normally open position and verify that STANDBY legend of S4 is illuminated.

4.7.3.1.4 Output voltages (J1). Apply voltages to J8 as specified in 4.7.3 plus item 3 and 4 as indicated in Figure 1. With STANDBY legend of S4 illuminated in compliance with conditions stated in 3.5.1.4, verify, using item 6, that the following voltages are available at J1:

a. 28 ±0.1 Vdc between J1-1(+) and J1-2(Ret).

b. 28 ±0.1 Vdc between J1-3(+) and J1-2(Ret).

c. 28 ±0.1 Vdc between J1-5(+) and J1-6(Ret).

d. 33 ±0.1 Vdc between J1-7(-) and J1-8(Ret).

e. 0 to NMT 7.4 Vdc, variable by operation of AGC ADJUST R1, between J1-A2(+) and J1-4(Ret).

4.7.3.1.5 RADIATE/STANDBY switch (S5)

4.7.3.1.5.1 STANDBY legend. Apply voltages to J8 as specified in 4.7.3. Under conditions stated in 3.5.1.5.1, verify that STANDBY legend of S5 is illuminated.

CAUTION - HIGH VOLTAGE

4.7.3.1.5.2 STANDBY mode. Connect P2 and J5. Refer to schematic 9333080 and connect voltmeter, item 10, across C8 positive lead to junction of R3 and C8. Set up conditions as stated in 3.5.1.5.2. Set S10 to TEST "A" position. Close circuit breaker CB1. After 20 ±2 seconds verify using item 10, that voltage across C8 is NLT 4.0 and NMT 5.0 kVdc.

4.7.3.1.5.3 RADIATE legend and radiate mode. Remove all input voltages and use shorting bar to remove residual voltages from

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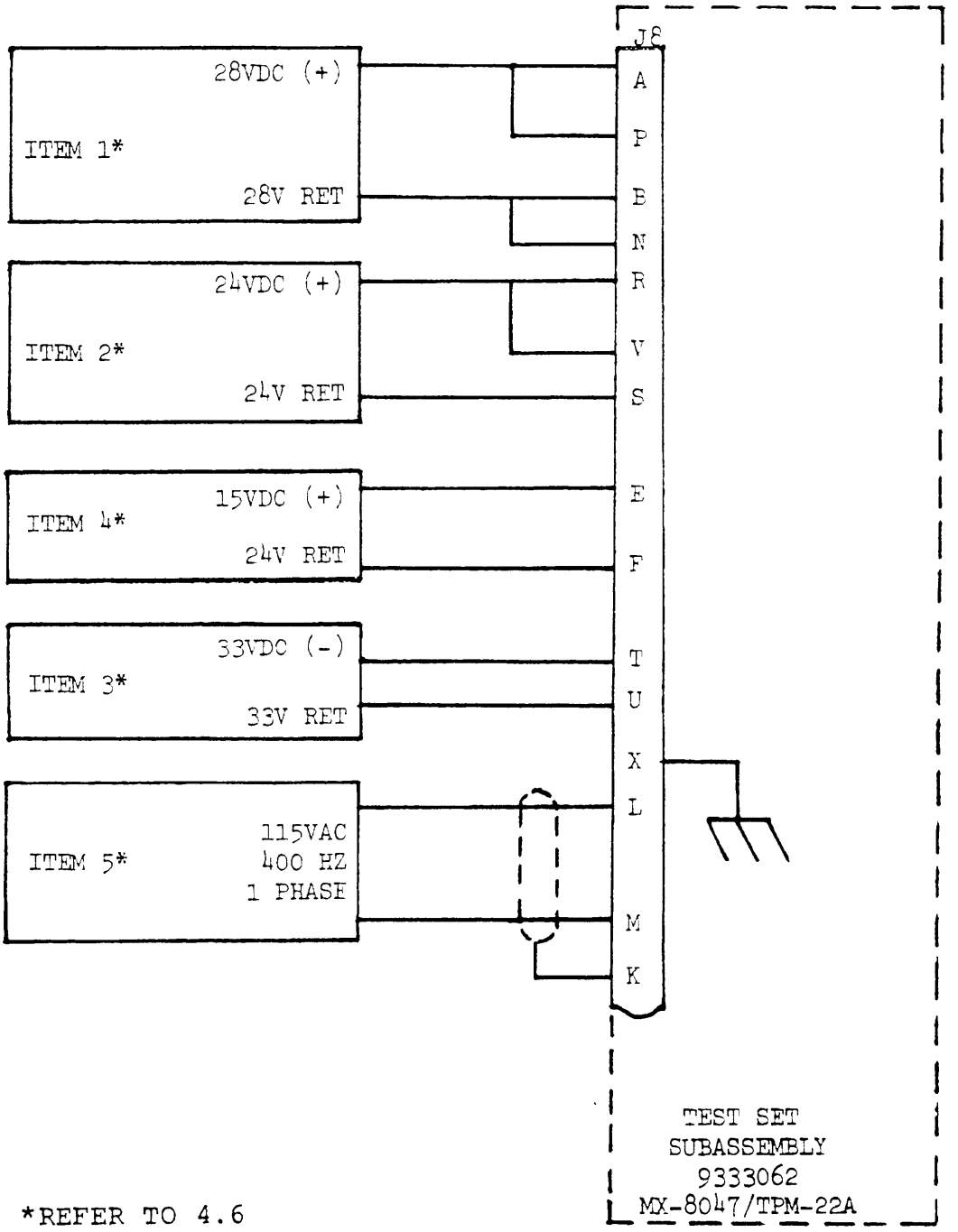


Figure 1. Power supply input test setup

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high voltage points. Connect P2 into J5. Close circuit breaker CB1. Connect a shorting wire between K5 - normally open and normally closed contacts. Using item 11 apply a pulse to J9 with the following characteristics:

width	=	0.5 usec
amplitude	=	20 ± 2 volts peak
repetition rate	=	20 kHz ±1 kHz
rise time	=	less than 0.05 nsec

reference: MOD trigger pulse, Figure 2.

Set S10 to TEST "B" position. Connect oscilloscope, item 12, with High Voltage probe, item 13, to junction of C8 and P2-E. Using MOD Trigger Pulse as scope sync apply voltages to J8 as specified in 4.7.3 and set up conditions in 3.5.1.4. Verify that STANDBY legend of S5 is illuminated. After 2 minutes ±12 seconds, operate S5 to normally open position and verify that the following occur:

a. RADIATE legend of S5 is illuminated.

b. STANDBY legend of S5 is extinguished.

c. Pulse displayed on oscilloscope, item 12 conforms to characteristics in Figure 2. After completion of test, disconnect P2 from J5.

4.7.3.1.6 KLYSTRON FILAMENT CURRENT meter (M1). Set up conditions as specified in 3.5.1.6.2 and S10 to TEST "A" position. Verify that initial indication of M1 is 1.5 amperes ±10%. Verify that after 20 ±2 seconds, meter M1 indication is NLT 2.0 and NMT 2.8A.

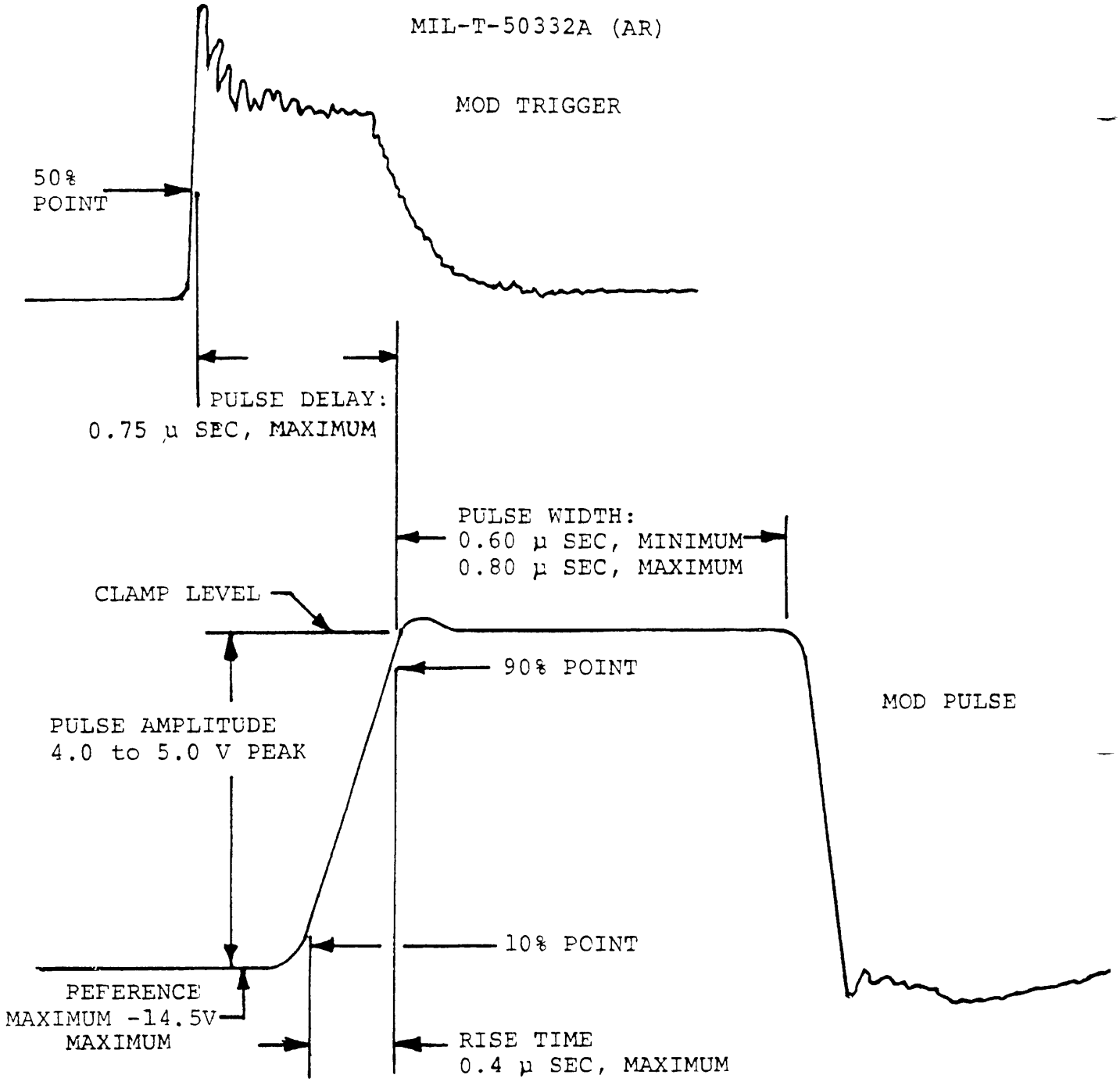
4.7.3.1.7 2A1 OVERLOAD indicator (S6). Disconnect P1 from subassembly A1.

4.7.3.1.7.1 High voltage power supply overload. Apply 28 ±0.1 Vdc between P1-1(+) and chassis ground (Ret). Verify that HIGH VOLTAGE POWER SUPPLY legend of S6 is illuminated.

4.7.3.1.7.2 Modulator overload. Apply 28 ±0.1 Vdc between P1-3(+) and chassis ground (Ret). Verify that MODULATOR legend of S6 is illuminated.

4.7.3.1.8 2A1 OVERLOAD reset switch (S6). Apply voltages to J8 as specified in 4.7.3. With STANDBY legend of S5 illuminated in compliance with conditions stated in 3.5.1.3.2, actuate and hold S6 in the normally open position and verify that 28 ±0.1

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*THE MOD PULSE AMPLITUDE MUST BE EQUAL OR GREATER THAN THE DC BIAS SO THAT THE CLAMP LEVEL IS AT ZERO VOLTAGE LEVEL.

Figure 2. Mod pulse, 1A1A2A1, output waveform

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Vdc is available between P1-2 (+) and chassis ground (Ret) . Use item 6 to measure voltage.

4.7.3.2 KLYSTRON POWER SUPPLY TEST section 2A2

4.7.3.2.1 Interlock indicator lamp (DS7) . Apply voltages to J8 as specified in 4.7.3. Under conditions stated in 3.5.2.1a. through g. , verify that continuity exists between J8-S and TB2-4. If any one of the conditions is not met., verify that continuity does not exist. Verify that DS7 is illuminated when S9 is pulled out in manual position and extinguished when S9 is pushed in or returned to its center (OFF) position.

4.7.3.2.2 POWER switch (S1)

4.7.3.2.2.1 OFF legend. Apply voltages to J8 as specified in 4.7.3. Under conditions as stated in 3.5.2.2.1, verify that OFF legend of S1 is illuminated.

4.7.3.2.2.2 STANDBY Legend. Apply voltages to J8 as specified in 4.7.3. Under condition as stated in 3.5.2.2.2, operate S1 to the normally open position and verify that STANDBY legend of S1 is illuminated. Verify that 28 ±0.1 Vdc is available between J3-A3 (+) and J3-A2 (Ret). Use item 6 to measure voltage.

4.7.3.2.3 OPERATE/STANDBY switch (S2).

4.7.3.2.3.1 STANDBY legend and standby mode. Apply voltages to J8 as specified in 4.7.3. Under conditions stated in 3.5.2.2.2, operate S2 to the normally closed position and verify that the STANDBY legend of S2 is illuminated.

4.7.3.2.3.2 OPERATE legend and operate mode. Under conditions listed in a. through e. below, verify that conditions listed in f. through h. exist.

a. Voltages applied to J8 as specified in 4.7.3.

b. Controls and conditions met in accordance with 3.5.2.2.2.

c. switch S2 set to normally open position.

d. Apply MOD TRIGGER pulse (Figure 2) to J9 using item 11.

e. Switch S10 set to “B” position.

f. OPERATE legend of S2 illuminated.

9. 28 ±0.1 Vdc available between J3-A3 (+) , J3-A4(+) and J3-A2 (Ret)

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h. 9.0 $^{+0.5}_{-0}$ kVdc available between J2 (-) and chassis ground using item 10.

i. 185 volt peak-to-peak symmetrical square wave is measured between J3-13 and J3-14 using item 12.

4.7.3.2.4 KLYSTRON POWER SUPPLY CURRENT meter (M2)

4.7.3.2.4.l Standby current. Apply voltages to J8 as specified in 4.7.3. Apply a resistive load of 14 ohms ±1%, 100 watts, item 8 between J3-A3 and J3-A2. With conditions as stated in 3.5.2.2.2, verify that meter M2 indicates 2.0 amperes ±6%.

4.7.3.2.4.2 Operate current. Apply voltages to J8 as specified in 4.7.3. Apply a resistive load of 14 ohms ±1%, 100 watts, item 8 between J3-A3 and J3-A2. Apply an identical load between J3-A4 and J3-A5. With conditions as stated in 3.5.2.4.2 (S2 OPERATE legend illuminated) , verify that meter M2 indicates 4.0 amperes ±6%.

4.7.3.2.5 2A2 OVERLOAD. indicator (S3)

4.7.3.2.5.1 High voltage power supply overload. Using item 1, apply 28 ±0.1 Vdc to J3-1(+) and J3-16 (Ret). Verify that HIGH VOGLTAGE POWER SUPPLY legend of S3 is illuminated.

4.7.3.2.5.2 Modulator overload. Using item 1, apply 28 ±0.1 Vdc to J3-3(+) and J13-16 (Ret). Verify that MODULATOR legend of S3 is illuminated.

4.7.3.2.6 2A2 OVERLOAD reset switch (S3). Apply voltages to J8 as specified in 4.7.3. With conditions as stated in 3.5.2.2.2, 3.5.2.3.1 or 3.5.2.3.2, actuate and hold S3 in the normally open position. Verify that 28 ±0.1 Vdc is available between J3-2(+) and chassis ground (Ret) . Use item 6 to measure voltage.

4.7.3.2.7 TEST A/B switch S10. Apply voltages to J8 as specified in 4.7.3. Operate S10 to TEST A position and set up conditions as stated in 3.5.2.3.2a. through c. Verify using item 9 that continuity exists between P2-C and chassis ground.

4.7.3.2.8 Voltage divider (A3). Connect the HV probe, item 13, to the input of A3. Connect the output of the item 13 to Channel 1 of the oscilloscope, item 12. Connect J6 to Channel 2.

a. Set up conditions as specified in 3.5.1.5.3. Note that the signal displayed on the oscilloscope shall have the same DC bias, amplitude, and shape. Set the oscilloscope as follows:

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Oscilloscope Setting

Coupling - DC
Sensitivity - 5v/div
Sweep 10 sec/div
Sweep mode - alternate

b. Should the signal on Channel 2 be different from the signal on Channel 1, adjust C1 of A3 until the signals have the same DC bias, amplitude, and shape.

4.7.3.2.9 EXTERNAL BEAM POWER SUPPLY test setup. Connect 100 ohm load resistor, item 14, between J11-C and J11-B, connect oscilloscope, item 12, with HV probe, item 13, across load resistor, item 14, connect shorting wire, item 15, between TP6 and TP7, connect shorting wire, item 16 between J11-E and J11-A and with conditions as specified in 4.7.3.1.3.2 use oscilloscope to determine compliance with 3.5.2.9.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet 9333062.

6. NOTES

6.1 Intended use. This high voltage tester is part of the AN/TPM-22A Radar Test Set, which is used for troubleshooting and performance checking of specific weapon system components.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this and the item detail specifications.

b. Applicable stock number.

c. A block of serial numbers to identify each test set.

d. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

e. First Article Sample requirements if other than specified in Section 4 of item detail specification.

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f. Packaging requirements , if other than specified in Section 5 of item detail specification.

g. Certificate of conformance for each lot or shipment of product.

6.2.2 Contract data requirements.

a. Contract data requirements for inspection equipment designs conforming to Data Item Description DI-R-1714 tailored, with appropriate addendum to each. See 6.3.

b. Data Cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410.

6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government designs and contractor designs. Government designs are designated by drawing numbers in the appropriate paragraph in. Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government design is provided; however, unless otherwise specified, the, contractor may submit alternate or modified contractor designs in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 Government designs. Government designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in DOD-STD-100.

c. An envelope drawing, as defined in DOD-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which Government designs are not specified and include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient infor-

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mation to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with Government designs if not otherwise covered thereby (see 6.3.1c) . Designs shall be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423) . The item detail specification number, paragraph number and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies,

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAF-I (D) , Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawinqs prepared by and identified as US Army Armament Research and Development Command, (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal. Technical data originally prepared by these activities is now under the, cognizance of ARDC.

6.5 Changes to previous issue Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-50332A	TEST SET, SUBASSEMBLY MX-8047/TPM-22A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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